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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 7.4

$$\begin{aligned}
 & \int \frac{\sqrt[5]{x^3} + \sqrt[6]{x^5}}{\sqrt[4]{x}} dx \\
 &= \int \frac{\sqrt[5]{x^3}}{\sqrt[4]{x}} dx + \int \frac{\sqrt[6]{x^5}}{\sqrt[4]{x}} dx \\
 &= \int x^{\frac{3}{5} \cdot \frac{-1}{4}} dx + \int x^{\frac{5}{6} \cdot \frac{-1}{4}} dx \\
 &= \int x^{\frac{12}{20} \cdot \frac{-5}{20}} dx + \int x^{\frac{10}{12} \cdot \frac{-3}{12}} dx \\
 &= \int x^{\frac{7}{20}} dx + \int x^{\frac{7}{12}} dx \\
 &= \frac{x^{\frac{27}{20}}}{\frac{27}{20}} + \frac{x^{\frac{19}{12}}}{\frac{19}{12}} + c \\
 &= \frac{20 \sqrt[20]{x^{27}}}{27} + \frac{12 \sqrt[12]{x^{19}}}{19} + c \\
 &= \frac{20x \sqrt[20]{x^7}}{27} + \frac{12x \sqrt[12]{x^7}}{19} + c
 \end{aligned}$$

References